

Yufeng Chen

(+1) 765-775-9220 | chen4044@purdue.edu | yufchen.github.io

Education

Ph.D. Candidate, Computer Science

Purdue University, West Lafayette, Indiana

Sep. 2021 – Expected Aug. 2027

3.83/4.0

Bachelor of Engineering, Electronic Information Engineering

University of Science and Technology of China, Hefei, China

Talent Program in Information Science and Technology

Sep. 2017 – Jul. 2021

GPA: 3.87/4.3

Research & Engineering Projects

Efficient Streaming of Immersive Content over Wi-Fi Networks

2022 – Present

Ph.D. Thesis Research, Purdue University

- Developed cross-layer scheduling frameworks for networked VR that coordinate application utility and transport behavior to improve perceptual quality under bandwidth and loss constraints.
- Designed packet-level progressive streaming for static 3D/VR assets and evaluated QUIC/SCTP policies, including stream prioritization and out-of-order delivery, to reduce application-level blocking.
- Developed adaptive policies that balance new data, retransmissions, and random linear network coding redundancy for fair, robust multi-user VR over lossy Wi-Fi.
- Explored joint scheduling of static assets and delay-sensitive real-time updates, including synchronization, consistency, and interaction traffic.

QoE-based Routing Path Selection

2022 – 2023

Purdue University and Juniper Networks

- Investigated routing path selection driven by application-level quality of experience rather than only traditional network-layer metrics.
- Contributed to the design and evaluation of QoE-aware path-selection strategies for networked applications.

Scheduling Diverse QUIC Streams

2022

Purdue University

- Extended quiche to study sender-side scheduling across reliable QUIC streams and unreliable datagrams for applications with mixed reliability and latency requirements.
- Implemented static, dynamic-priority, and weighted round-robin schedulers; evaluated CPU, memory, and queueing-delay tradeoffs.

Hierarchical Multi-Domain Optical Network Testbed with DeepRMSA

Jun. 2020 – Nov. 2020

University of California, Davis

- Extended DeepRMSA from a single-domain simulator to a hierarchical multi-domain RMSA system, coordinating broker- and domain-level DRL agents for intra- and inter-domain connection requests.
- Implemented an ONOS-based multi-campus testbed with three Java controllers and Python/TensorFlow actor-critic agents; processed 1,965,300 requests in an 80-hour experiment with 148.87 ms average inter-domain request latency.

RDMA Optimization for MongoDB

Jun. 2018 – Nov. 2018

University of Science and Technology of China

- Replaced TCP/IP-based MongoDB cluster data transfer with RDMA to reduce client-server and inter-server communication latency.
- Achieved up to $3.58\times/6.16\times$ peak Put/Get throughput improvement over the TCP/IP baseline.

Publications

- Yuqi Zhou, Shuqi Liao, **Yufeng Chen**, Sonia Fahmy, and Voicu Popescu. “Scalable Collocated Multi-User VR Through Virtual Environment User Spatial Coherence.” *IEEE Transactions on Visualization and Computer Graphics*, 2026. Also in Proc. IEEE VR, Mar. 2026.
- **Yufeng Chen**, Umakant Kulkarni, Voicu Popescu, and Sonia Fahmy. “RUN: A Case for Cross-Layer Networked Virtual Reality.” *Proceedings of the 33rd ACM International Conference on Multimedia*, MM '25, pp. 12111–12120, Oct. 2025.
- Umakant Kulkarni, **Yufeng Chen**, Patrick Melampy, and Sonia Fahmy. “Toward QoE-based Routing Path Selection.” *IEEE HPSR*, pp. 114–119, Jun. 2023.
- **Yufeng Chen**, Akhil Prasad, Sonia Fahmy, and Voicu Popescu. “Scheduling Diverse QUIC Streams.” Poster, USENIX NSDI, Apr. 2023.

Teaching Experience

- Teaching Assistant, CS 536: Data Communication and Computer Networks 2024, 2025
- Teaching Assistant, CS 240: Programming in C Spring 2022
- Teaching Assistant, CS 251: Data Structures Fall 2021

Technical Skills

Programming: Python, C/C++, Java, Shell

Networking & Systems: QUIC, TCP/IP, Wi-Fi, multicast/broadcast, socket programming, Linux, RDMA

VR & Graphics: Unity, point clouds, 3D Gaussian Splatting

Tools: Git, Docker, Wireshark, ns-3, MATLAB